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Bioavailability and Pharmacokinetics of Oxazepam

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Summary. Six healthy volunteers received oxazepam 15 mg i.v. and orally at an interval of at least one week. The kinetic variables of i.v. oxazepam were: elimination half-life ($t_{1/2\beta}$) 6.7 h, total clearance (CL) 1.07 $ml \cdot min^{-1} \cdot kg^{-1}$, volume of distribution (V_c) 0.27 $l \cdot kg^{-1}$ (0.21-0.49) and volume of distribution at steady-state (V_{ss}) 0.59 $l \cdot kg^{-1}$. The intravenous disposition of unbound oxazepam was characterized by a clearance of 22.5 $ml \cdot min^{-1} \cdot kg^{-1}$ and a distribution volume of 12.3 $l \cdot kg^{-1}$. After oral oxazepam the peak plasma level was reached in 1.7 to 2.8 h. The plasma $t_{1/2\beta}$ at 5.8 h was not significantly different from the i.v. value. Absorption was almost complete, with a bioavailability of 92.8%. Urinary recovery was 80.0 and 71.4% of the dose after intravenous and oral administration, respectively. Renal clearance (CL_R) of the glucuronide metabolite was 1.10 $ml \cdot min^{-1} \cdot kg^{-1}$ (0.98-1.52). Oxazepam was extensively bound to plasma protein with a free fraction of 4.5%.

Key words: oxazepam; pharmacokinetics, i.v.-/oral administration, bioavailability, healthy volunteers

Oxazepam, a 3-hydroxy-1,4-benzodiazepine derivative, which is almost exclusively metabolised by glucuronide conjugation [1, 2] and may serve therefore as a suitable tool for investigating Phase II drug metabolism. The pharmacokinetics of oral oxazepam has been described by several workers [1-7]. However, the lack of a preparation for intravenous administration has prevented precise estimation of the pharmacokinetic variables, especially the clearance, volume of distribution and systemic availability [8].

The present authors prepared a solution of oxazepam for intravenous administration and have performed a study to: (1) determine the bioavailability of oxazepam; (2) measure its pharmacokinetic variables after i.v. administration; and (3) characterize the renal excretion pattern of the glucuronide metabolite.

Materials and Methods

Six healthy female and male subjects gave their informed consent to participation in the study. They ranged in age from 26 to 38 years (Table 1). All

Table 1. Clinical and pharmacokinetic data for intravenous oxazepam 15 mg in 6 subjects

Subject	Sex F/M	Age (years)	Body weight (kg)	V_c ($l \cdot kg^{-1}$)	V_{ss} ($l \cdot kg^{-1}$)	$t_{1/2\alpha}$ (min)	$t_{1/2\beta}$ (h)	AUC ($ng \cdot ml^{-1} \cdot min$)	Total clearance ($ml \cdot min^{-1} \cdot kg^{-1}$)	48 h Excretion of oxazepam glucuronide % of dose	$t_{1/2}$ Determined from oxazepam glucuronide excretion rate vs time (h)
1	F	27	58	0.21	0.37	4.9	5.5	338.4	0.76	75.3	7.2
2	M	32	65	0.28	0.77	17.8	6.6	161.3	1.43	79.3	7.8
3	M	38	68	0.25	0.56	3.5	5.9	203.0	1.09	81.3	6.0
4	M	38	87	0.39	0.88	16.4	6.8	119.9	1.44	74.7	5.7
5	F	29	56	0.49	0.62	32.9	6.9	258.5	1.04	78.7	5.9
6	M	26	70	0.22	0.42	18.3	9.2	398.3	0.54	85.3	9.4
Median				0.27	0.59	17.1	6.74	230.8	1.07	79.0	6.6

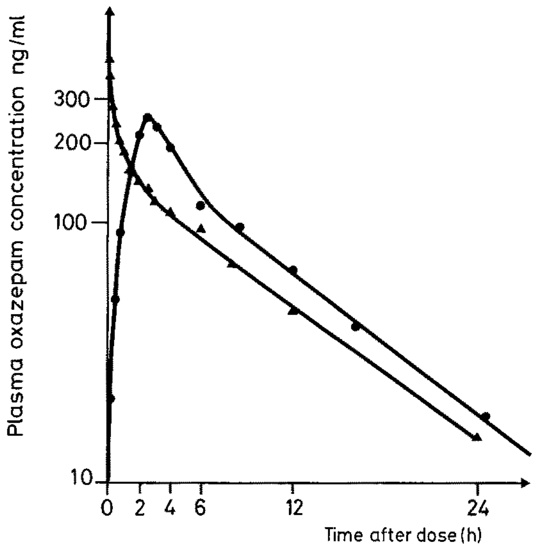

Fig. 1. Plasma oxazepam concentrations after intravenous (▲) and oral (●) administration of 15 mg oxazepam to Subject 4

were non-smokers, who only drank alcohol socially, and had taken no drugs during the preceding four weeks. The protocol was approved by the local Ethics Committee.

The subjects received in random order 15 mg oxazepam i.v. and orally, after an overnight fast, on 2 days separated by at least 1 week. For intravenous administration a solution of oxazepam (10 ml containing 15 mg oxazepam, 4 g propylene glycol, 1.5 g ethanol and 3.8 g sterile water) was infused over a period of 15 min. Oral oxazepam was administered as 15-mg tablets (Serepax, Ferrosan) together with 100 ml tap water. The subjects continued to fast for a further 2 h after dosing.

Venous blood samples were collected in heparinized tubes before, 15, 30, 45, 60, 90 min, and 2, 3, 4, 6, 8, 10, 12, 16 and 24 h after drug administration. Following the i.v. dose additional samples were collected 2 and 5 min post-infusion.

Urine was collected 0–4, 4–8, 8–24 and 24–48 h after the dose. Plasma samples and aliquots of all urine collections were stored at $-20°$ until analyzed.

Analysis of Plasma and Urine

Unchanged oxazepam and total drug in plasma and urine after enzymatic hydrolysis were determined by a modified HPLC method [9]. Pure oxazepam was kindly provided by Ferrosan A/S.

The method was modified as follows: The plasma sample 1 ml was mixed with 1 N NaOH 50 μl and 10 μl internal standard solution (100 μM diazepam) and extracted with 4 ml absolute ether for 10 min. The organic phase was separated and evaporated to dryness over nitrogen at 60°C. The residue was reconstituted with 100 μl mobile phase (methanol/distilled water, 60/40; v/v). 25 μl was injected into an HPLC system consisting of a Perkin-Elmer automatic injector ISS 100, Series 10 isocratic pump, variable wavelength UV-detector LC-85 set at 230 nm, automatic integrator LCI 100, and a 12.5 cm column, i.d. 4.6 mm, packed with Spherisorb ODS 5 μ by a dilute slurry technique. The system was supplied with a 3.5 cm precolumn packed with Spherisorb RP18 30–40 μ. The flow rate was $1.5\,\mathrm{ml \cdot min^{-1}}$. Enzymatic deconjugation of plasma and urine samples was performed by incubating overnight at 37°C 0.5 ml sample with 0.5 ml sodium acetate 0.1 M pH 4.5 containing 25 μl glucuronidase/aryl sulphatase ($100.000\,\mathrm{IU \cdot ml^{-1}}$, Boehringer) and 7 N NaOH 20 μl. The samples were thereafter analyzed by the same procedure as described for unchanged oxazepam.

The detection limit was oxazepam 5–10 ng/ml plasma. Interassay variation of identical samples did not exceed 7%.

Plasma protein binding of oxazepam in each subject was determined in a single nonheparinized, drug free serum sample to which $1.0\,\mathrm{\mu g \cdot ml^{-1}}$ oxazepam was added, according to the detection limit of the analysis. Within the usual range of therapeutic doses of oxazepam, concentration-dependent plasma protein binding should not be suspected of significantly altering the pharmacokinetics [10].

The extent of binding was determined by equilibrium dialysis [11], as modified by Christensen et al. [12]. The samples were dialyzed for 3 h at 37°C. The surface to volume relationship in the system was sufficiently large for that period to suffice to obtain equilibrium.

Analysis of Data

Plasma oxazepam concentrations were analyzed by the ESTRIP program for kinetic studies [13]. According to the best fit, the data could be described by an open two-compartment model. Correction of the relevant terms for the infusion period was done subsequently [14].

The area under the 24 h plasma-concentration curve (AUC) was calculated by the trapezoidal approximation. The residual area extrapolated to in-

Table 2. Pharmacokinetics of oral oxazepam 15 mg in 6 subjects

Subject	lag time (min)	Peak conc. ($ng \cdot ml^{-1}$)	Time to peak h	$t_{1/2\alpha}$ min	$t_{1/2\beta}$ h	AUC ($ng \cdot ml^{-1} \cdot min$)	F %	48 h Excretion of oxazepam glucuronide % of dose	$t_{1/2}$ Determined from oxazepam glucuronide excretion rate vs time; h
1	5.1	567	2.5	20	5.7	322.8	95.4	78.7	10.1
2	4.2	294	2.0	36	6.0	139.0	86.2	68.7	5.3
3	12.7	514	2.0	40	5.4	226.4	111.5	72.7	7.2
4	17.7	257	2.5	48	5.9	129.9	108.3	70.0	5.2
5	4.9	408	1.7	36	5.6	226.7	87.7	66.0	7.6
6	0	405	2.8	60	8.4	358.9	90.1	75.3	7.6
Median	5.0	407	2.3	38	5.8	226.6	92.8	71.9	7.4

Table 3. Intravenous disposition of unbound oxazepam and renal clearance of its glucuronide metabolite

Subject	Unbound drug %	Clearance ($ml \cdot min^{-1} \cdot kg^{-1}$)	Volume of distribution ($l \cdot kg^{-1}$)	Renal clearance of glucuronide CL_R (4-8 h) $ml \cdot min^{-1} \cdot kg^{-1}$
1	4.8	15.8	7.7	1.12
2	3.8	37.6	20.3	1.52
3	4.2	26.0	13.3	1.30
4	3.7	38.9	23.8	1.08
5	5.5	18.9	11.3	1.05
6	4.8	11.3	8.8	0.98
Median	4.5	22.5	12.3	1.10

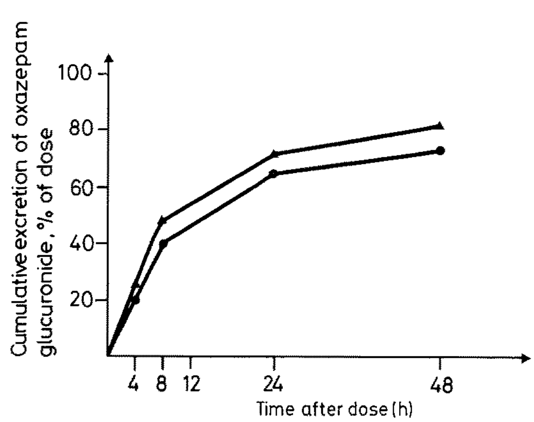

Fig. 2. Cumulative urinary excretion of oxazepam glucuronide following intravenous (▲) and oral (●) administration of 15 mg oxazepam over 48 h. Each point is the mean of 6 subjects

finity was calculated as the 24-h concentration value divided by β. Total clearance was calculated as dose divided by AUC. The systemic availability of oral oxazepam was calculated by dividing the oral by the intravenous AUC. The hybrid parameters from the fitted function after the i.v. dose were used to calculate the volume of the central compartment and the volume of distribution at steady-state (V_{ss}), the distribution half-life ($t_{1/2\alpha}$) and elimination half-life ($t_{1/2\beta}$) according to standard procedures [14]. When necessary, the function after the oral dose was modified by the introduction of a lag time, which, together with the absorption half-life ($t_{1/2\alpha}$) was estimated by the ESTRIP program. The elimination half-life of oxazepam glucuronide in urine was determined from the excretion rate vs time data. Renal clearance of oxazepam glucuronide over the time interval 4-8 h after i.v. administration of oxazepam was calculated by dividing the amount of glucuronide in urine by the average glucuronide concentration in plasma over the same period of time.

A two-tailed Wilcoxon rank sum test was used for statistical analysis; $p < 0.05$ was chosen as level of significance.

Results

Representative i.v. and oral plasma curves are shown in Fig. 1. After i.v. administration the initial distribution phase proceeded rapidly, with a half-life of less than 20 min in all but one subject (Table 1). After oral administration absorption proceeded with a half-life ranging from 20 to 60 min, and the peak plasma concentration was reached within 2.8 h (Table 2). The bioavailability of oral oxazepam was 92.8% (86.2-111.5, median and range). The elimination half-lives after i.v. and oral oxazepam were 6.7 h (5.5-9.2) and 5.8 h (5.4-8.4), respectively (median, range). The values did not differ significantly. The i.v. clearance was 1.07 $ml \cdot min^{-1} \cdot kg^{-1}$ (0.54-1.44; median, range). Volume of the central compartment (V_c) was 0.27 ($l \cdot kg^{-1}$ (0.21-0.49). Distribution volume at steady-state (V_{ss}) was slightly less than body weight, with a median value of 0.59 and a range from 0.37 to 0.88 $l \cdot kg^{-1}$. The corresponding value for unbound clearance was 22.5 $ml \cdot min^{-1} \cdot kg^{-1}$ (11.3-38.9, median, range) and for unbound volume of distribution 12.3 $l \cdot kg^{-1}$ (7.7-23.8), respectively (Table 3).

Protein binding of oxazepam was extensive with a free fraction of 4.5% (3.7-5.5, median, range). The 48 h excretion of intact oxazepam did not exceed 1% of the dose in any of the subjects. 79.0% (74.7-85.3) and 71.9% (66.0-78.7, median, range) of the dose could be accounted for in the urine after intravenous and oral dosing (Fig.2). The renal clearance of oxazepam glucuronide after i.v. administration was $1.10\,\mathrm{ml \cdot min^{-1} \cdot kg^{-1}}$ (0.98-1.52; Table 3). The half-life determined from the oxazepam glucuronide excretion rate vs time data were 6.6 h (5.7-9.4) after i.v. and 7.4 h (5.2-10.1) after po administration.

Subjective CNS effects appeared soon after i.v. administration and mainly consisted of sedation and drowsiness. The symptoms subsided within 1 to 2 h. The same effects were seen within 1 to 2 h after oral administration, but they were less pronounced. Four of six subjects had pain during the i.v. infusion and 2 of them developed a clinically overt, mild phlebitis, from which they eventually recovered. Propylene glycol has recently been incriminated as the cause of lactic acidosis in isolated cases [15]. No clinical symptoms suggestive of this diagnosis were seen here.

Discussion

The kinetics of oxazepam could readily be described by an open two-compartment model. This had previously been demonstrated in humans [3, 8, 9], but the lack of a suitable intravenous preparation had made estimation of the pharmacokinetic variables dependent solely on the oral administration of oxazepam. The present study is the first to employ intravenous administration of oxazepam in humans. The preparation was made for experimental purposes only, since both the composition and the amount of the vehicle required for administration of a therapeutic dose would preclude its commercial applicability. The distribution of oxazepam following intravenous infusion proceeded rapidly, with a mean half-life of 17.1 min, corresponding to the fast onset of sedation. The volume of distribution suggested moderately extensive penetration into tissues. The elimination half-life and clearance of total and unbound oxazepam were very similar to values reported earlier [3, 8, 9]. Also in accordance with those studies, the major metabolic pathway appeared to be conjugation to the glucuronide. More than 70% of the dose was recovered in urine as the glucuronide within 48 h of drug administration. The high renal clearance of oxazepam glucuronide was compatible with the fact that most glucuronides have a high renal extraction ratio. In a recent study a comparable clearance was found for the glucuronide [16].

In previous pharmacokinetic studies of oxazepam, estimates of the volume of distribution and total systemic clearance from appropriate functions have been based on the assumption of 100% systemic availability of orally administered oxazepam, since no preparation for intravenous use was available. The present study has demonstrated that this assumption was quite realistic, since the absorption of oral oxazepam appeared to be more than 90% complete. The actual values rule out failure of a significant fraction of the orally administered drug to reach the systemic circulation due to first-pass metabolism. The level of the systemic clearance also classifies oxazepam as a drug with a low extraction ratio [17].

The slower and lesser pronounced pharmacodynamic effects of oxazepam after oral administration could be ascribed to the slow absorption process. It has been suggested that a rapid increase in concentration is more important than the absolute level achieved [18].

In conclusion, the results substantiate the assumption that the systemic availability of oral oxazepam is close to 100%. It is considered valid, therefore, to use oral administration of oxazepam in future pharmacokinetic studies of factors affecting the glucuronidation pathway in humans.

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References

1. Greenblatt DJ, Shader RI, Koch-Weser J (1975) Pharmacokinetics in clinical medicine. Oxazepam versus other benzodiazepines. Dis Nerv Syst 36: (no. 5, section 2) 6-13
2. Chinnery R, Sundwall A (1978) Pharmacodynamic, pharmacokinetic and clinical aspects on oxazepam and related benzodiazepines. Acta Psychiatr Scand 274 [Suppl]: 1-128
3. Wretlind M, Pilbrant A, Sundwall A, Vessman J (1977) Disposition of three benzodiazepines after single oral administration in man. Acta Pharmacol Toxicol 40 [Suppl 1]: 28-39
4. Greenblatt DJ (1981) Clinical pharmacokinetics of oxazepam and lorazepam. Clin Pharmacokinet 6: 89-105
5. Ochs HR, Greenblatt DJ, Otten H (1981) Disposition of oxazepam in relation to age, sex, and cigarette smoking. Klin Wochenschr 59: 899-903
6. Murray TG, Chiang ST, Koepke HH, Walker BR (1981) Renal disease, age, and oxazepam kinetics. Clin Pharmacol Ther 30: 805-809
7. Greenblatt DJ, Murray TG, Audet PR, Locniskar A, Koepke HH, Walker BR (1983) Multiple-dose kinetics and

dialyzability of oxazepam in renal insufficiency. Nephron 34: 234-238
8. Greenblatt DJ, Divoll M, Harmatz JS, Shader RI (1980) Oxazepam kinetics: Effects of age and sex. J Pharmacol Exp Ther 215: 86-91
9. Patwardhan RV, Yarborough GW, Desmond PV, Johnson RF, Schenker S, Speeg KV (1980) Cimetidine spares the glucuronidation of lorazepam and oxazepam. Gastroenterology 79: 912-916
10. Moschitto LJ, Greenblatt DJ (1983) Concentration-independent plasma protein binding of benzodiazepines. J Pharm Pharmacol 35: 179-180
11. Weder HG, Schildknecht J, Kesselring P (1971) A new equilibrium dialyzing system. Am Lab 10: 15-21
12. Christensen JH, Andreasen F, Jensen EB (1980) The binding of thiopental to serum proteins determined by ultrafiltration and equilibrium dialysis. Acta Pharmacol Toxicol 47: 24-32
13. Brown DR, Manno JE (1978) ESTRIP, a BASIC computer program for obtaining initial polyexponential parameter estimates. J Pharm Sci 67: 1687-1691
14. Gibaldi M, Perrier D (1975) Drugs and the pharmaceutical sciences, Vol 1: Pharmacokinetics. Marcel Dekker, New York
15. Kelner MJ, Bailey DN (1985) Propylene glycol as a cause of lactic acidosis. J Anal Toxicol 9: 40-42
16. Van Hecken AM, Tjandramaga B, Verbesselt R, De Schepper PJ (1985) The influence of diflunisal on the pharmacokinetics of oxazepam. Br J Clin Pharmacol 20: 225-234
17. Wilkinson GR, Shand DG (1975) A physiological approach to drug clearance. Clin Pharmacol Ther 18: 377-390
18. Bliding A (1974) Effects of different rates of absorption of two benzodiazepines on subjective and objective parameters. J Clin Pharmacol 7: 201-211

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